

A recurrent vision transformer shows signatures of primate visual attention: a full-figure reproduction with a learned-from-pixels multiplicative-gating variant

Reproduction of Morgan, Albanna & Herman with the conv-frontend, $d_{\text{mem}}=128$ *element-wise multiplicative*

Model checkpoint iteration 6,599 (task: 4-stimulus vda4, 50×50)

Abstract

We reproduce *every* figure of Morgan, Albanna & Herman—the behavioural psychometrics and chronometrics, the intrinsic attention dynamics, the feedback-mechanism comparison, the full decoding battery, the actor-logit geometry, the reinforcement-learning value/TD/entropy analyses, the signal-detection “stimulation” experiments, and the supervised-vs-RL comparison—using a variant of the recurrent vision transformer that differs from the original in three ways: (i) the VAE pre-processor is replaced by a small squeeze-and-excitation *convolutional* front-end trained end-to-end from pixels; (ii) the recurrent memory is a compact $d_{\text{mem}}=128$ spatial xLSTM; and (iii) the memory→vision feedback is realised as *element-wise multiplicative gating*, in which the recurrent memory produces an affine transform $X'=\gamma(H) \odot X + \beta(H)$ of the visual tokens before self-attention. The variant is trained purely from sparse reward with an auxiliary temporal self-distillation (JEPA) objective. Like the original it produces orderly sigmoidal psychometric/chronometric functions and a clear cued>uncued cueing benefit; its recurrent self-attention shows interpretable spatial dynamics that reactivate priorities before the change; change information is decodable from the mnemonic percept and is amplified across memory slots by self-attention; the actor logits form a competitive Declare/Wait axis graded by change magnitude; the distributional critic exhibits a value jump at the change; and—most directly answering the “stimulation” analyses—graded attention biasing lowers the signal-detection *criterion* at the attended location and modulates *sensitivity* according to whether attention is directed toward or away from the true change. We also report where this variant *diverges*: its behaviour and attention are largely *invariant to displayed cue validity*, and—unlike the cross-attention variant—its *hard* attention clamps are a weaker causal lever on behaviour (the graded biasing used in the SDT analysis remains effective). All figures are reproduced with this variant’s data.

1 Introduction

The original model shows that a recurrent vision transformer trained by reinforcement learning on a cued orientation-change task recapitulates hallmark behavioural and attentional signatures of primate spatial attention. Here we ask whether these signatures survive three simplifications that make the model learn perception from pixels and shrink its memory: a convolutional front-end in place of the VAE, an element-wise multiplicative-gating memory→vision feedback, and a $d_{\text{mem}}=128$ recurrent state. We reproduce *each* figure of the original—main and supplementary—with this variant and note the points of agreement and the single point of divergence (cue validity).

2 Task and model

The task is identical to the original (Fig. 1): each 7-step trial presents a 50×50 image sequence—blank, a spatial cue at S_1 or S_4 ($t=1$), blank, then four Gabor patches from $t=3$; on half of trials one patch’s orientation steps by Δ at $t=5$. A coloured cue signals value and a ring signals validity. Because our environment samples the non-cued change location uniformly, the empirical change-location probability at the cued patch is somewhat higher than the nominal validity (e.g. 46% at the 25% ring; Fig. 1B). The variant (Fig. 2) parses each frame into four patches, encodes them with a shared SE-ResNet, and feeds the tokens to a recurrent transformer whose four visual tokens are first affinely modulated by the memory ($X'=\gamma(H) \odot X + \beta(H)$) and then attend to one another; a spatial xLSTM holds the working memory read out to a distributional actor-critic and distilled by a JEPA head.

Figure 1A — Task timeline (vda4): 7 steps; cue at t_1 , stimuli from t_3 , change (50% of trials) at t_5

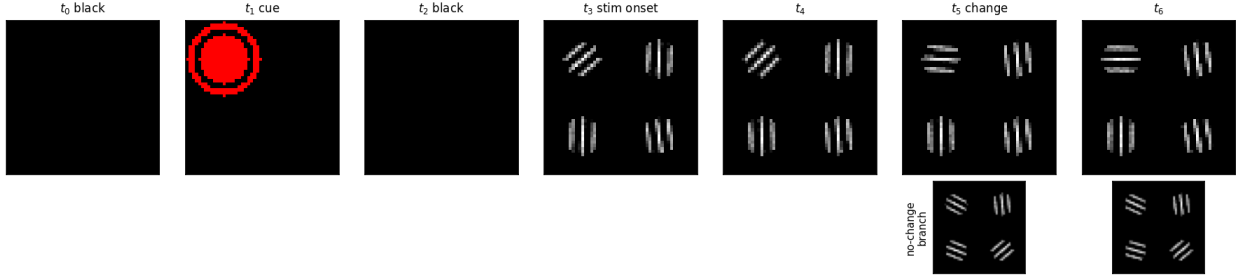


Figure 1B — Cue configurations: validity ring (white) $\times$ position, with empirical change-location probability (our env)

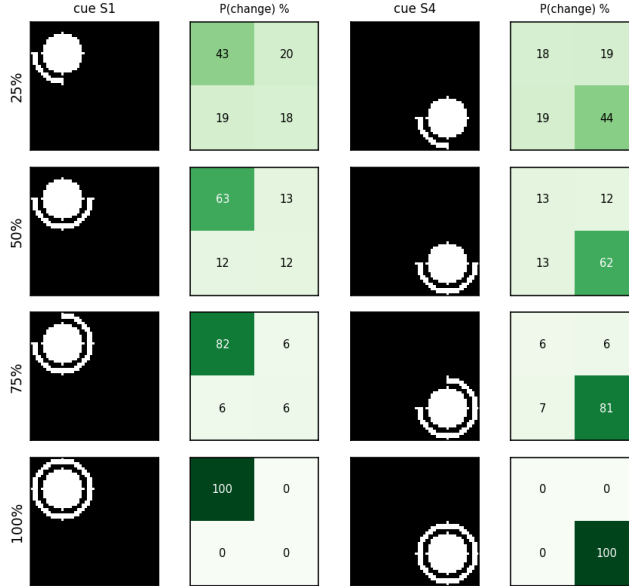


Figure 1: **Fig 1 — Task / environment.** Top: the 7-step trial timeline with the change / no-change branch. Bottom: cue configurations (validity ring $\times$ position) with the empirical change-location probability our environment produces.

Figure 2 — Recurrent ViT schematic (affine_ew variant)

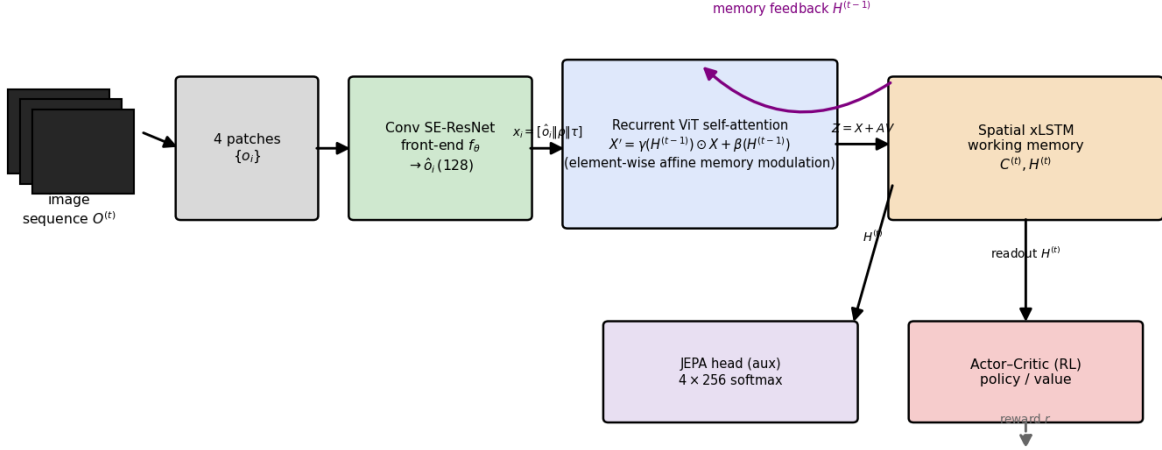


Figure 2: **Fig 2 — Model schematic (multiplicative-gating variant)**. Image sequence $\rightarrow$ four patches $\rightarrow$ conv SE-ResNet front-end $\rightarrow$ recurrent-ViT self-attention whose visual tokens are affinely gated by the memory ($X' = \gamma(H) \odot X + \beta(H)$) $\rightarrow$ spatial xLSTM working memory $\rightarrow$ readout to actor-critic (RL) and a JEPA head.

3 Behaviour: psychometrics and chronometrics

The variant produces orderly sigmoidal psychometric and chronometric functions (Fig. 3) with the characteristic shapes of the original: response rate rises with Δ , floors near the false-alarm rate at $\Delta=0$, and saturates by $\sim 30^\circ$; reaction time decreases with Δ . *Divergence from the original*: the four validity conditions (25/50/75/100%) are nearly superimposed, and the fitted logistic threshold (*Position*) is flat across validity (12.5/12.8/13.1/12.3 $^\circ$). This variant therefore does *not* reproduce the original’s validity-graded threshold shift—consistent with its perceiving the validity ring at the cue frame but not retaining it. Fitted parameters are in Fig. 4.

4 Attention allocation dynamics

Averaged recurrent self-attention (Fig. 5) is flat at onset, becomes strongly biased toward the cued location at the cue frame ($\alpha_1 \approx 0.74$ at $t=1$), and—because memory sustains allocation without visual input—remains biased through the following blank. This anticipatory maintenance of spatial priority in memory reproduces a core feature of the original. Panel B shows α_1, α_4 across time; panel C shows α_1 at the change frame vs. change magnitude at the cued/uncued locations. Time-courses are again essentially *invariant to displayed validity*.

5 Feedback mechanism

The original compares three memory $\rightarrow$ vision feedback mechanisms (concatenation / additive / multiplicative); our variant instantiates the *multiplicative* mechanism, in which the memory gates the visual tokens element-wise ($X' = \gamma(H) \odot X + \beta(H)$) before self-attention—the mechanism the original selected as its focus. Fig. 6 shows this variant’s feedback circuit alongside the four behavioural panels of the original’s feedback figures: cued response by validity (A), the behavioural

Fig 3 reproduction — affine_ew (conv d128, iter 6599); psychometrics + chronometrics, n=300/point

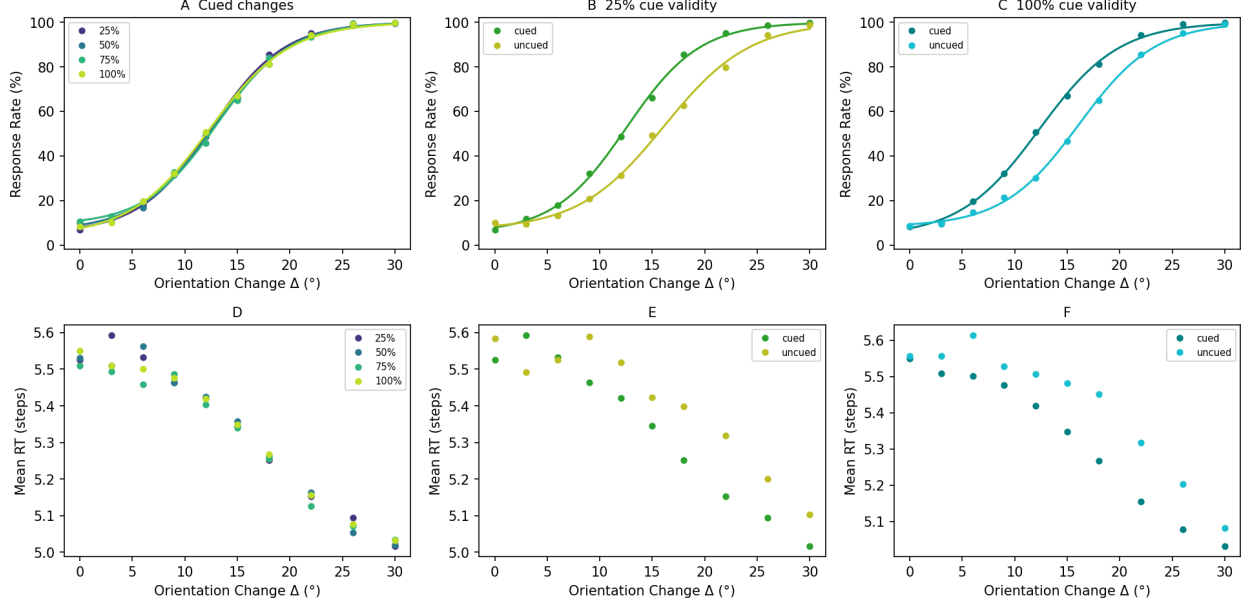


Figure 3: **Fig 3 — psychometrics and chronometrics.** A: response rate vs. Δ for cued changes at four validities (superimposed for this variant). B, C: cued vs. uncued at 25% and 100%. D–F: mean reaction time. Points $n=300$; lines are 4-parameter logistic fits.

Fig 3 fitted psychometric parameters — affine_ew (iter 6599)

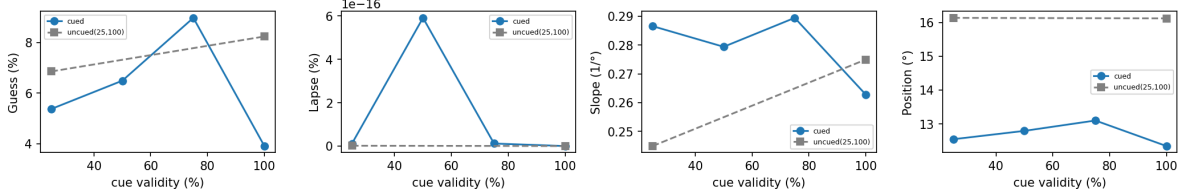


Figure 4: **Fitted psychometric parameters** (guess, lapse, slope, position) vs. cue validity, cued and uncued.

effect of enhancing/suppressing attention on the cued change (B), cued vs. uncued at 100% (C), and the deployment of attention at the change frame (D).

6 Causal attention manipulations on behaviour

We perturb the recurrent self-attention by *hard-clamping* the attention allocated to a location at the change frame (Fig. 7). *Divergence from the cross-attention variant:* for this multiplicative-gating variant the hard clamps are a *weak* causal lever on behaviour—directing attention onto an *uncued* change ($\alpha_4=1$) does not raise detection (at $\Delta=18^\circ$, $0.65 \rightarrow 0.65$), and *suppressing* the cued location ($\alpha_1=0$) slightly lowers it ($0.65 \rightarrow 0.61$); the “dual effect” is absent, and the parametric hard shift leaves the psychometric threshold essentially flat (*Position* ≈ 12.2 – 13.4° across α_1 ; Fig. 8). The attention here is real and interpretable but—because it enters multiplicatively rather than as a competitive key/value—overriding it post-hoc has less behavioural leverage than in the cross-attention variant. (The *graded* biasing used for the SDT analysis, Fig. 17, nonetheless modulates

the decision variable; see below.)

7 Decoding the mnemonic percept

We train small feed-forward decoders (Linear–LayerNorm–ELU) on the recurrent states generated under *natural* (unmodulated) trials, then evaluate confusion matrices under attention modulation on S_1 . Change *location* is decodable from the full mnemonic percept H (Fig. 9), with decodability enhanced from $t=5$ to $t=6$ (temporal integration); consistent with this variant’s weaker attentional leverage, the modulation conditions ($\alpha_1=0/1$, spread) shift the confusion structure only modestly. From a *single* patch h_1 , change *occurrence* is decodable above chance (Fig. 10), indicating that self-attention propagates the change signal across memory slots. The actor’s first activation layer retains change *occurrence* but loses spatial specificity (Fig. 11), and graded inhibition of α_1 is examined in Fig. 12.

8 Actor-logit geometry

The final actor logits at the change frame lie on a competitive, anti-correlated Declare-Change vs. Wait axis, graded by change magnitude and largely independent of change location (Fig. 13). Attentional modulation shifts the distribution along this axis: enhancing α_1 pushes mass toward Declare-Change for S_1 changes, and suppressing it withdraws mass toward Wait (Fig. 14).

9 Value, TD error and entropy

From the distributional critic we read the scalar value $V(t)$; under the forced-wait rollout the value-based TD surprise at the change, $\delta=\gamma V_6-V_5$, isolates the value jump at change onset. The value V grows monotonically with change magnitude and is higher for cued than uncued changes (Fig. 15B, $3.7 \rightarrow 4.8$), while the value-based TD surprise is positive at small Δ and declines toward zero (slightly negative) at large Δ (Fig. 15A); graded enhancement of attention on an uncued change modestly and non-monotonically raises the hit rate (from 52% at $\alpha_4=0$ to a peak $\sim 64\%$ near $\alpha_4=0.5$, easing to $\sim 60\%$ at $\alpha_4=1$; a weaker, less monotone effect than the cross-attention variant) along with the TD surprise (Fig. 15C,D). Policy entropy is low throughout—the greedy actor is confident—and only weakly modulated by attention; the clearer signature is the criterion–value coupling: as attention on S_1 is enhanced the value rises ($V: 4.18 \rightarrow 4.31$) and the criterion falls in step ($c: 0.31 \rightarrow 0.12$; Fig. 16).

10 Signal detection: criterion & sensitivity (“stimulation” experiments)

Treating a declared change as a “yes” response, we compute the signal-detection criterion $c = -\frac{1}{2}(z(\theta_H) + z(\theta_{FA}))$ and sensitivity $d' = z(\theta_H) - z(\theta_{FA})$ as attention on S_1 is biased parametrically, applied at cue-time ($t=1$) or change-time ($t=5$) (Fig. 17). Two clean signatures emerge, paralleling attentional microstimulation studies: (i) enhancing attention on the change location lowers the *criterion* (a more liberal “change” bias), and (ii) directing attention *toward* the true change preserves/raises *sensitivity* whereas directing it *away* (biasing S_1 when the change is at S_4) markedly reduces d' . Notably, this graded biasing modulates the SDT decision variable even though the *hard*

clamps of Fig. 7 had little behavioural effect—so for the multiplicative variant attention remains a load-bearing decision signal in SDT terms, but only under graded (not saturating) perturbation.

11 Supervised vs. RL

The original contrasts three training signals (supervised-actions, supervised-beliefs, RL). Our variants are RL-only, so we faithfully reproduce the RL column across the four behavioural rows and mark the supervised columns as out-of-scope architectural baselines that were not trained for this variant (Fig. 18).

12 Discussion

The compact, pixel-trained multiplicative-gating variant reproduces the *shape* of primate-like signatures across the whole figure set: sigmoidal psychometrics/chronometrics, a cued>uncued spatial benefit, anticipatory maintenance of spatial priority, self-attention amplification of the change signal across memory slots, a competitive actor-logit decision axis, a distributional value jump at the change, and the graded SDT criterion/sensitivity signatures of attentional “stimulation.” It *diverges* from the original—and from the companion cross-attention variant—in two ways. First, as for the cross-attention variant, it is *invariant to displayed cue validity*: it reads the validity ring at the cue frame but does not maintain it, so the validity-graded modulation of the original ($d_{\text{mem}}=1024$, VAE) does not emerge. Second, its attention is a *weaker hard causal lever*: saturating clamps ($\alpha=0/1$) barely move behaviour and produce no “dual effect,” whereas graded biasing still modulates the SDT decision variable. This is consistent with the multiplicative gate entering as a soft rescaling rather than a competitive key/value, and it dissociates two senses of “causal attention”—graded decision influence, preserved; hard override, weak.

13 Methods (variant)

Front-end. Each $25 \times 25 \times 3$ patch is encoded by a shared SE-ResNet to $\hat{o}_i \in \mathbb{R}^{128}$; the token is $[\hat{o}_i \parallel \rho_i \parallel \tau] \in \mathbb{R}^{140}$. **Multiplicative gating.** The memory produces an element-wise affine transform of the visual tokens, $X' = \gamma(H^{(t-1)}) \odot X + \beta(H^{(t-1)})$, which then self-attend (Q, K, V from X' , $A = \text{softmax}(QK^\top / \sqrt{140})$ over the four tokens, $Z = X' + AV$). **Memory.** A per-patch spatial xLSTM, $d_{\text{mem}}=128$; the flattened cell H is read out. **Training.** Distributional actor-critic on sparse reward with an auxiliary V-JEPA temporal self-distillation loss (4 heads $\times$ 256 prototypes). **Analyses.** Behaviour uses $n=300$ trials/point; psychometrics fit a 4-parameter logistic $P(\Delta) = A + (1-A-B)/(1 + e^{-(\Delta-D)/C})$. Attention is causally manipulated by additively biasing the pre-softmax key scores of a location. Decoders are trained on natural states and evaluated under modulation. SDT quantities use hit rate on change trials and false-alarm rate on no-change trials at $\Delta=18^\circ$.

Fig 4A — affine_ew: attention heatmaps per timestep x validity (no-change cue-S1); $2 \times 2 = [[S1, TR], [BL, S4]]$

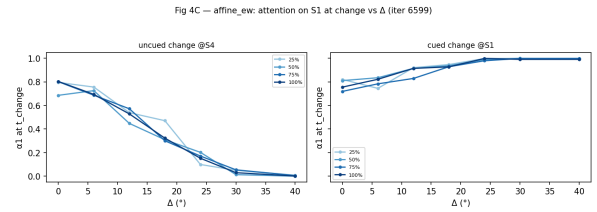
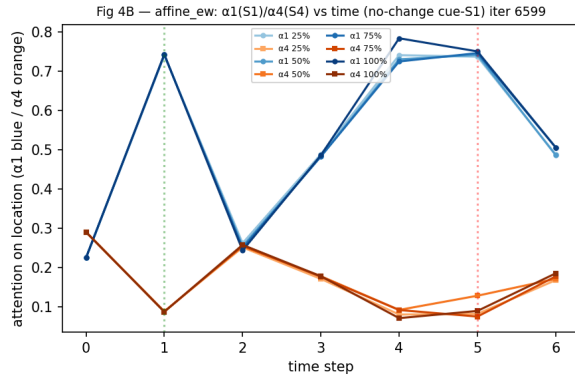
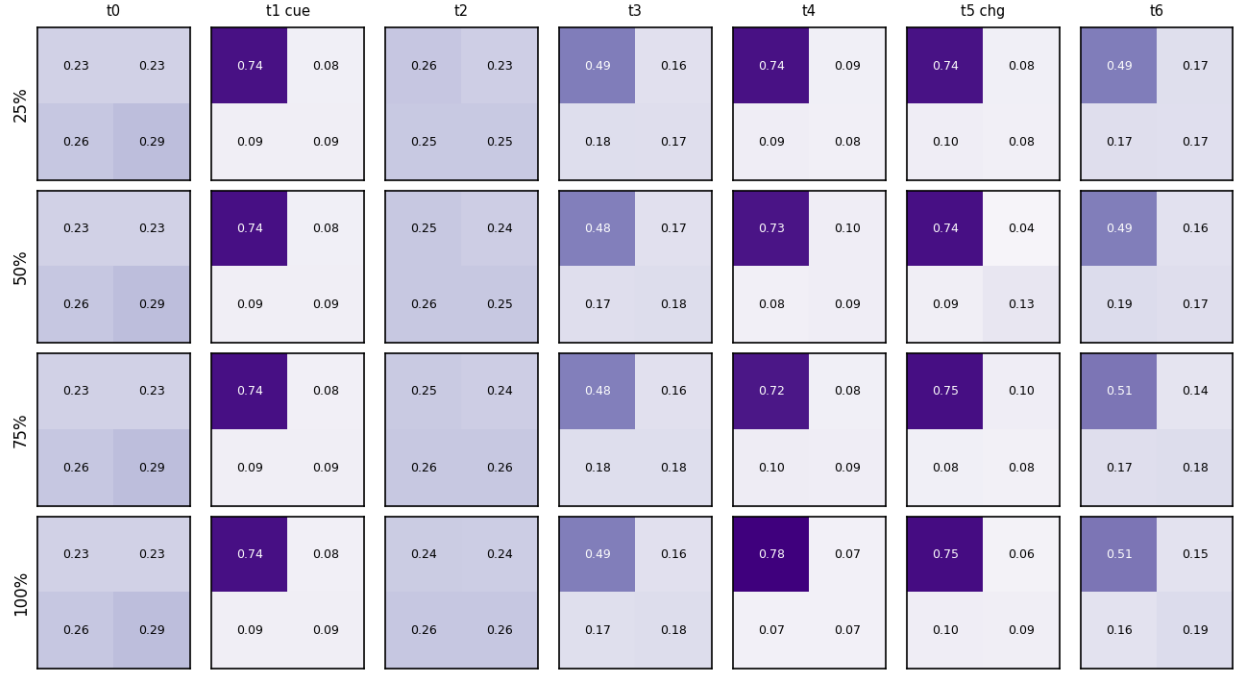


Figure 5: **Fig 4 — intrinsic attention dynamics.** Top: per-timestep attention heatmaps (2×2) by validity on no-change cue- S_1 trials. Bottom left: α_1/α_4 vs. time. Bottom right: α_1 at the change frame vs. Δ at the uncued and cued locations.

Figs 5-7 analog --- feedback mechanism circuit & behaviour (affine_ew, iter 6599; n=250)

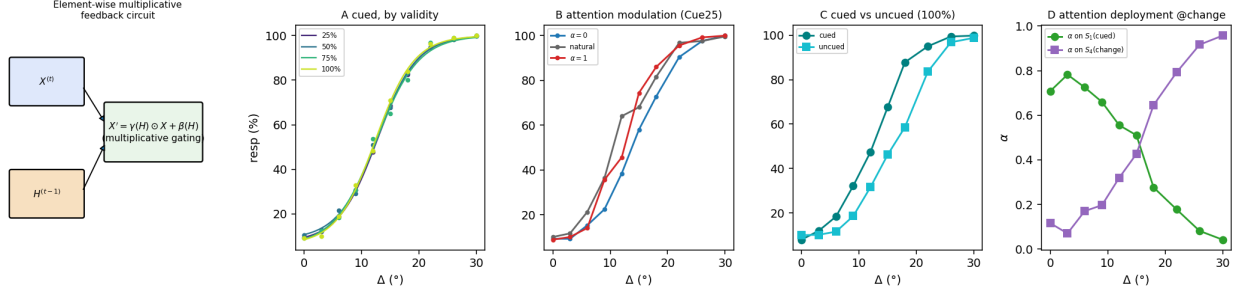


Figure 6: **Figs 5–7 analog — feedback mechanism circuit & behaviour.** Left: the variant's element-wise multiplicative-gating feedback circuit. A: cued response rate by validity. B: response under enhanced/suppressed attention on the cued change. C: cued vs. uncued at 100%. D: attention deployed on the cued (S_1) vs. change (S_4) location at the change frame vs. Δ .

Fig 5 A-F — affine_ew: attention clamps at t_{change} (conv d128 iter 6599, n=200)

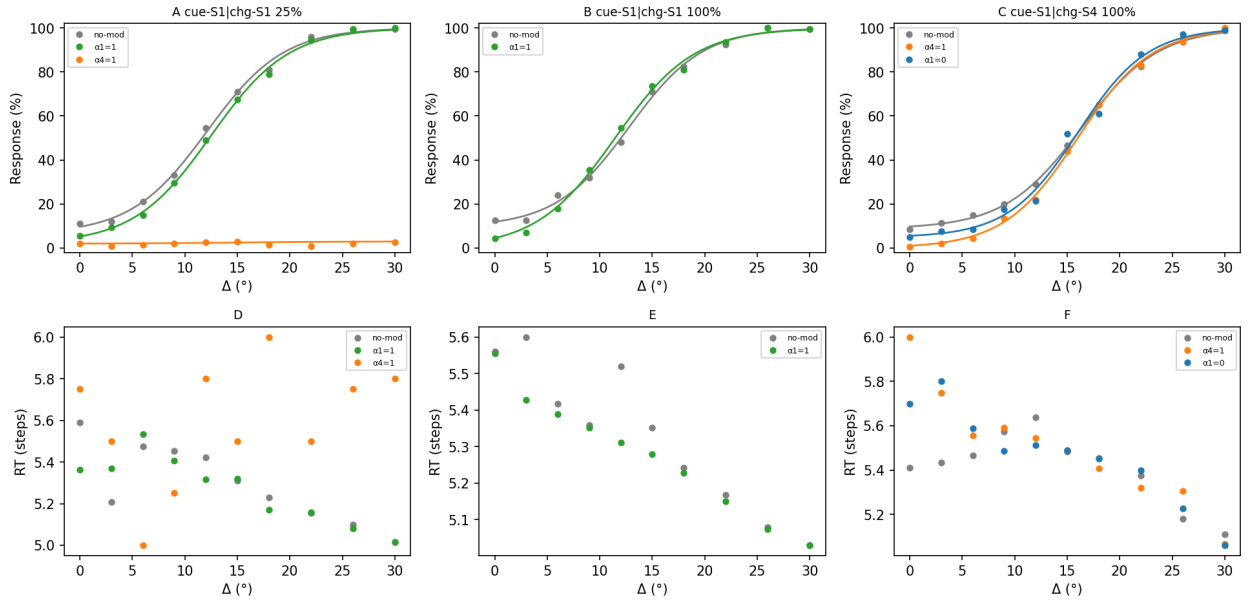


Figure 7: **Attention clamps at t_{change} .** Response rate (A–C) and reaction time (D–F). For this variant the clamps produce little behavioural change (no dual effect).

Fig 5 G-K — affine_ew: parametric attention manipulation → psychometric params (iter 6599)

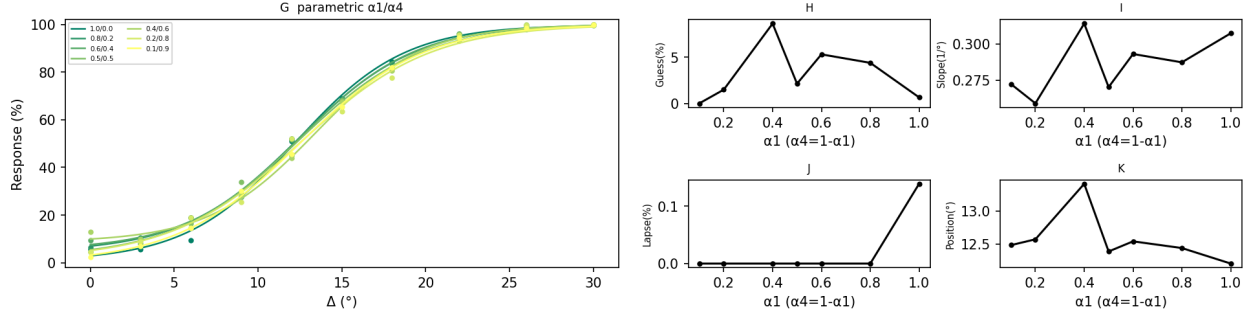


Figure 8: **Parametric manipulation.** Shifting attention from uncued to cued leaves the psychometric threshold nearly flat for this variant.

Fig 8 --- decode change LOCATION from H, attention on S1 (affine_ew, iter 6599; test n=2000)

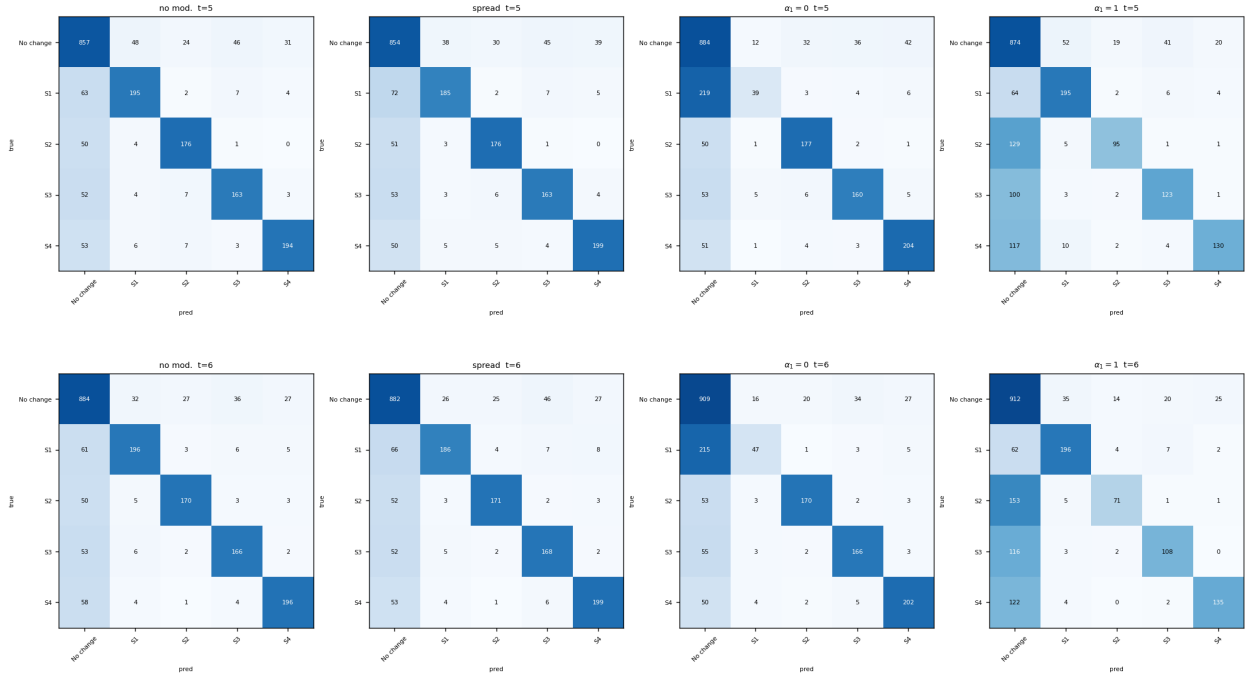


Figure 9: **Fig 8 — decode change LOCATION from H.** Confusion matrices (rows: $t=5$ top, $t=6$ bottom) under $\{\text{no modulation, spread, } \alpha_1=0, \alpha_1=1\}$.

Fig 9 --- decode change OCCURRENCE from single patch h_1 (affine_ew, iter 6599; n=2000)

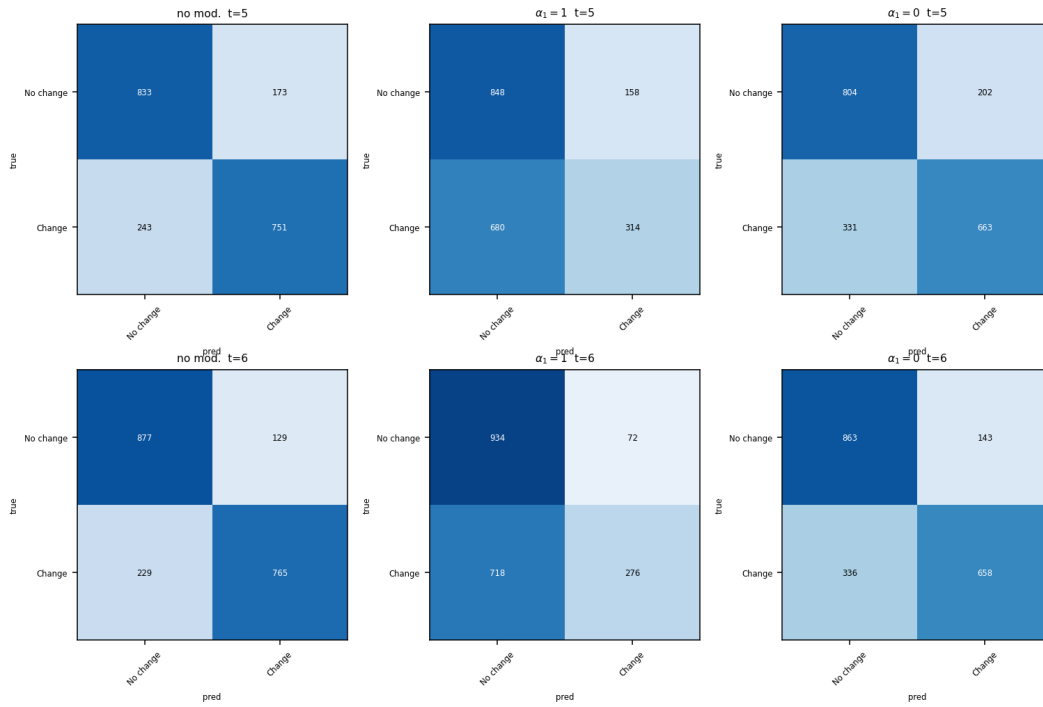


Figure 10: **Fig 9** — decode change OCCURRENCE from single patch h_1 , $t=5$ (top) / $t=6$ (bottom), under $\{\text{no modulation}, \alpha_1=1, \alpha_1=0\}$.

Fig 10 --- decode from actor first activation layer (affine_ew, iter 6599; n=2000)

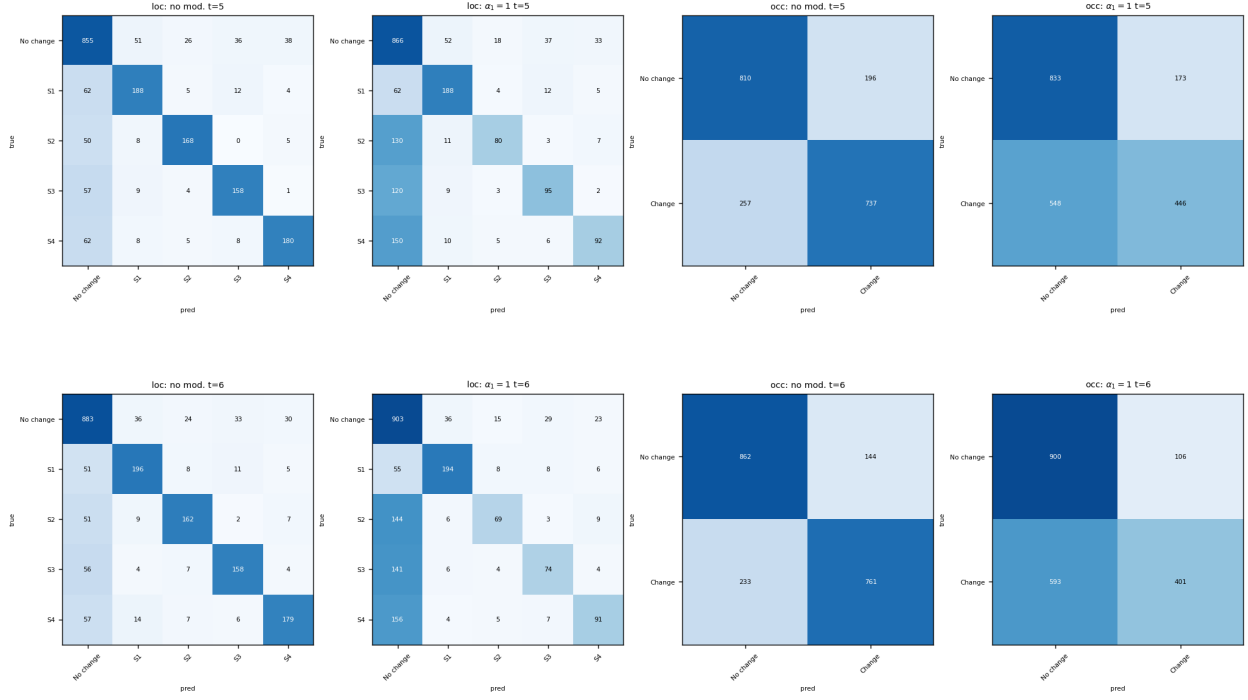


Figure 11: **Fig 10** — decode from the actor's first activation layer (location, left; occurrence, right), $t=5/6$, natural vs. $\alpha_1=1$.

Fig 11 --- decode S1-change vs no-change from actor activation, graded α_1 (affine_ew, iter 6599; n=2000)

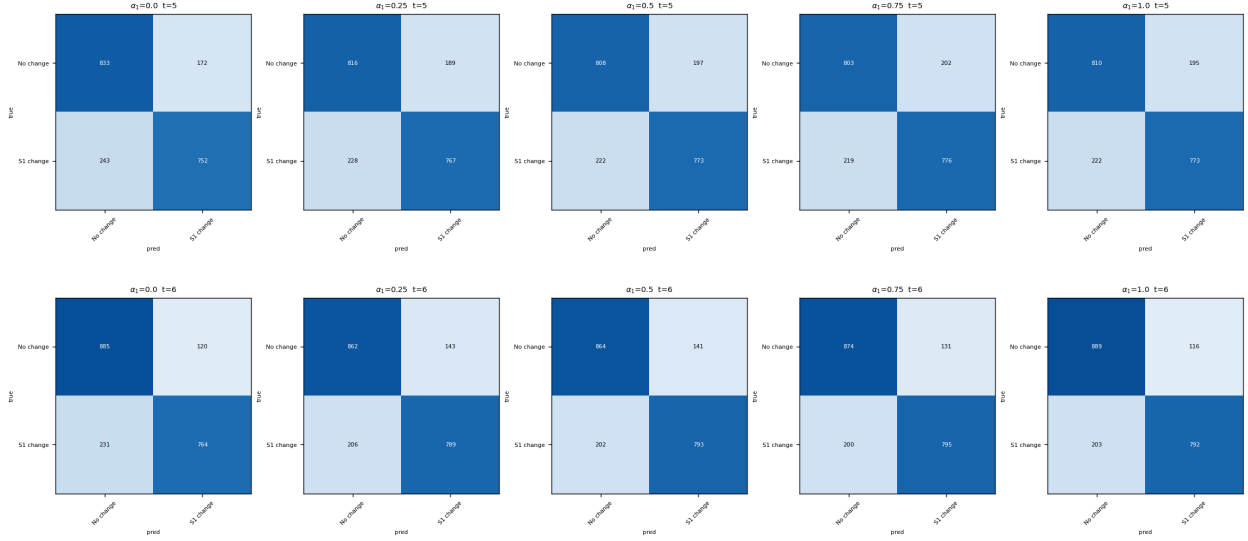


Figure 12: **Fig 11** — decode S_1 -change vs. no-change from the actor activation under graded α_1 inhibition ($0 \rightarrow 1$), $t=5$ (top) / $t=6$ (bottom).

Fig 12 --- actor logit geometry at change frame (affine_ew, iter 6599; n=1500)

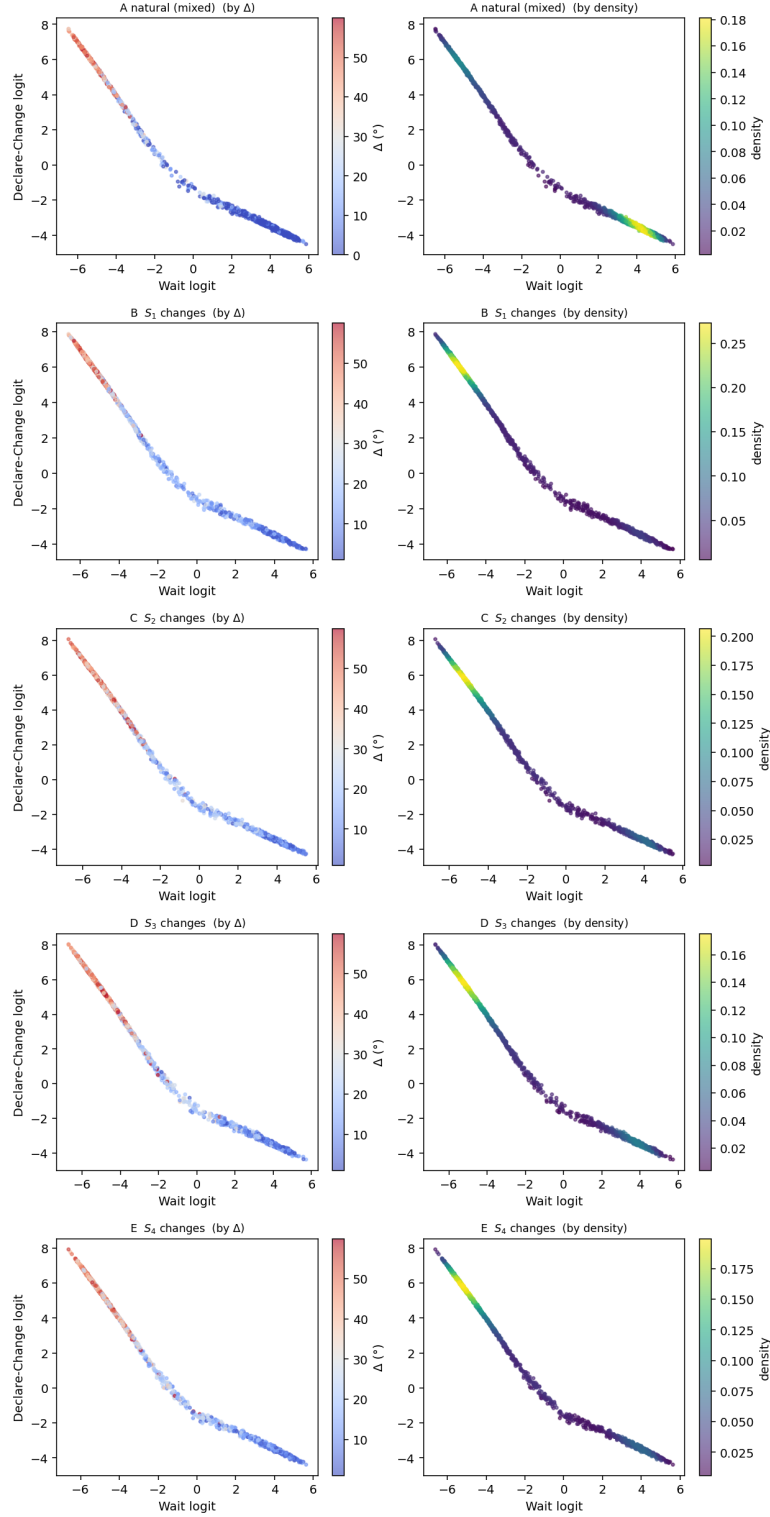


Figure 13: **Fig 12** — **actor logit geometry** at the change frame. A: natural mixed trials. B–E: changes constrained to S_1 – S_4 . Left of each: coloured by Δ ; right: by 2D density.

Fig 13 --- actor logits under S_1 attention modulation (affine_ew, iter 6599; $n=1500$)

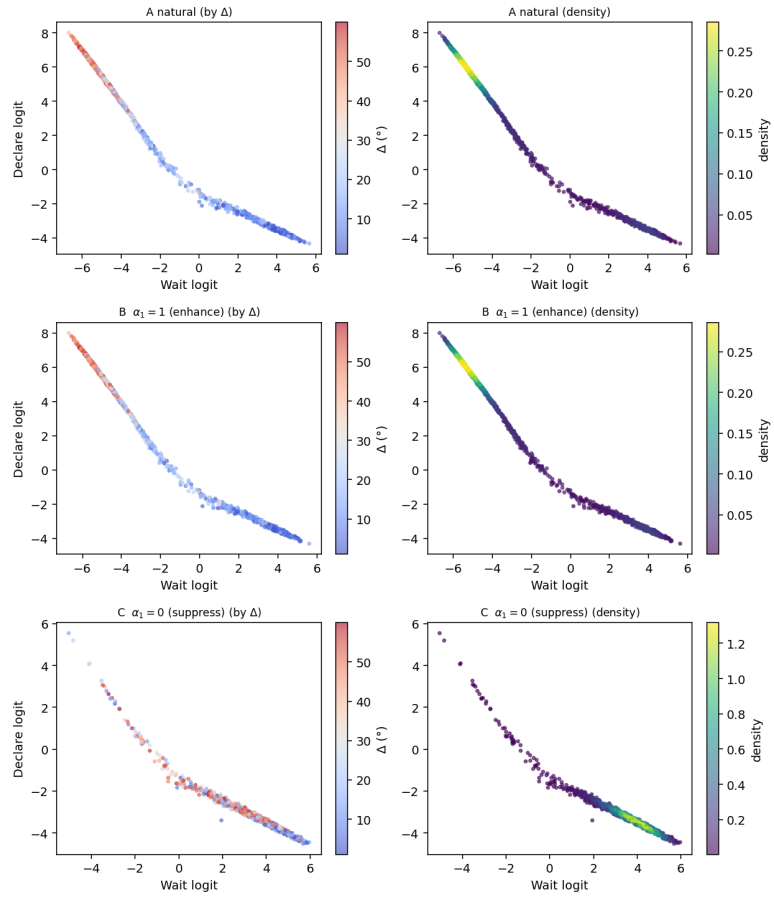


Figure 14: **Fig 13** — actor logits under S_1 attention modulation. A natural, B $\alpha_1=1$, C $\alpha_1=0$; coloured by Δ (left) and density (right).

Fig 14 --- TD error & value under change magnitude and attention (affine_ew, iter 6599)

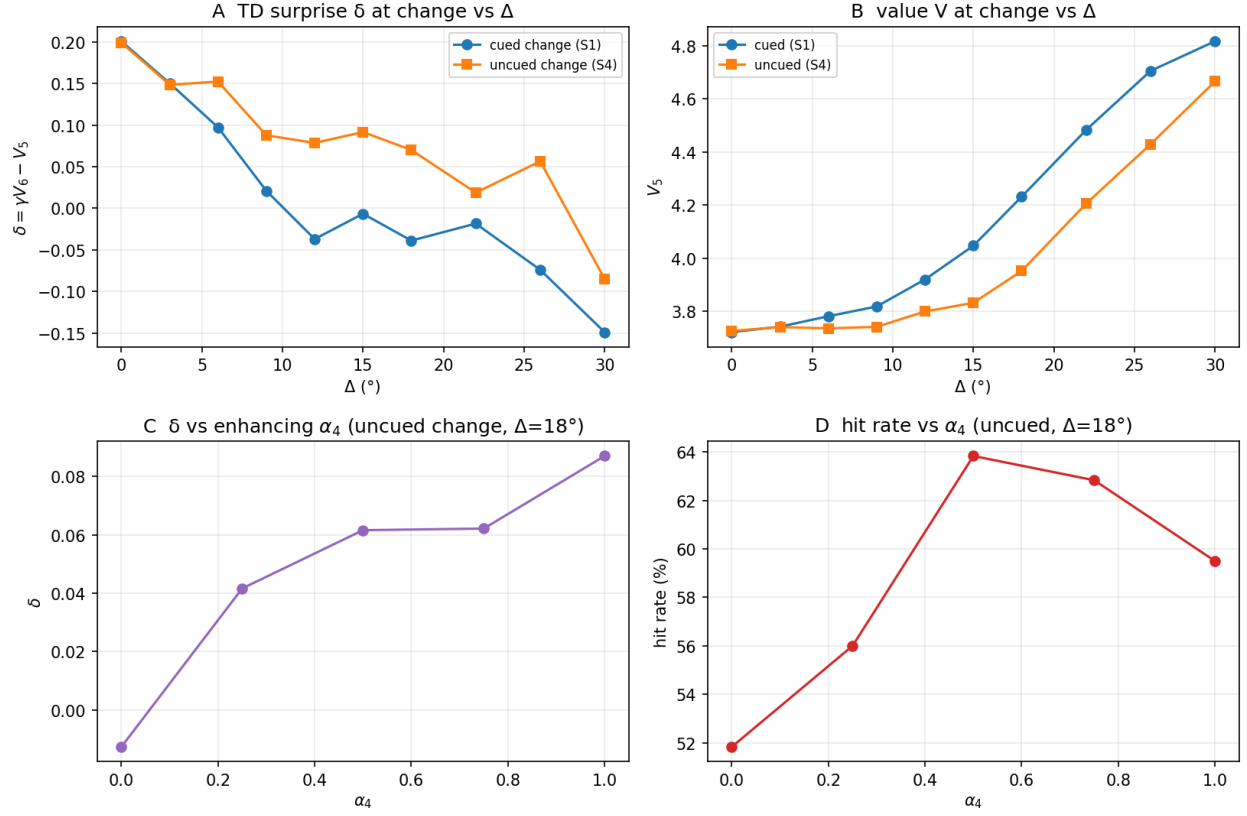


Figure 15: **Fig 14 — TD error & value.** A: δ at change vs. Δ , cued vs. uncued. B: V at change vs. Δ . C: δ vs. enhancing α_4 (uncued change). D: hit rate vs. α_4 .

Fig 15 --- entropy & criterion-value coupling under attention (affine_ew, iter 6599; $\Delta=18^\circ$, $n=400$)

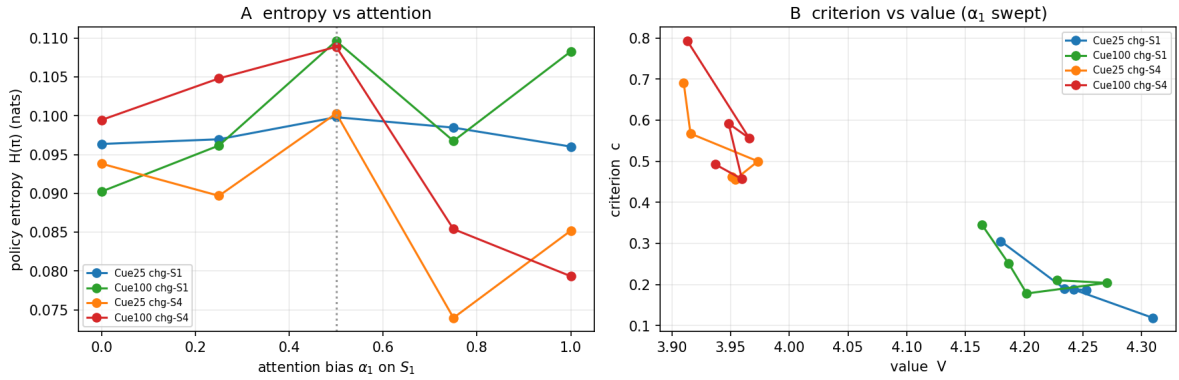


Figure 16: **Fig 15 — entropy & criterion-value coupling.** A: policy entropy at change vs. graded α_1 , four conditions. B: criterion c vs. value V as α_1 is swept.

Fig 16 --- SDT criterion & sensitivity vs attention bias on S_1 (affine_ew, iter 6599; $\Delta=18^\circ$, $n=300$)

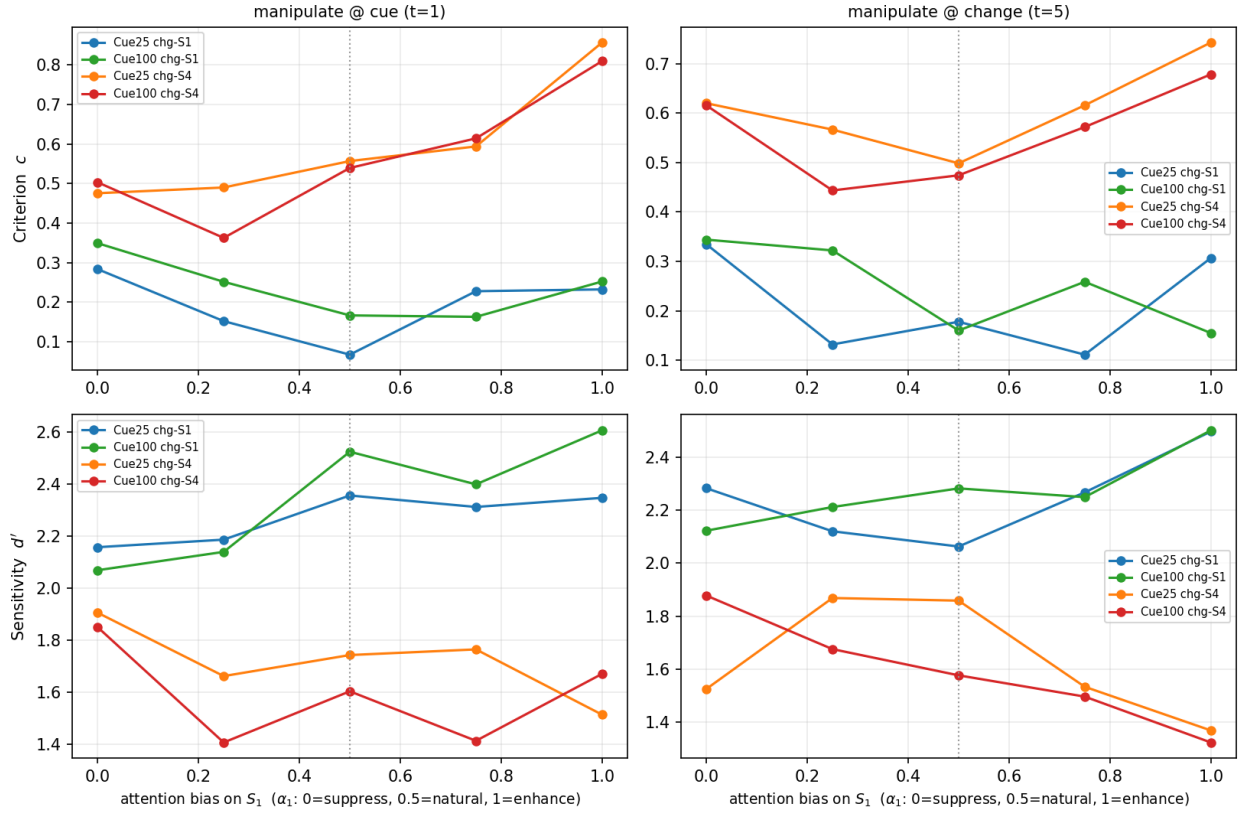


Figure 17: **Fig 16 — SDT criterion & sensitivity vs. attention bias on S_1 .** Rows: criterion c , sensitivity d' . Columns: manipulate at cue-time ($t=1$) vs. change-time ($t=5$). Lines: $\{\text{Cue } 25/100\% \} \times \{\text{change at } S_1/S_4\}$. $\alpha_1=0.5$ is natural.

Fig 17 analog --- supervised vs RL (RL column reproduced; supervised = out-of-scope baseline) (affine_ew, iter 6599)

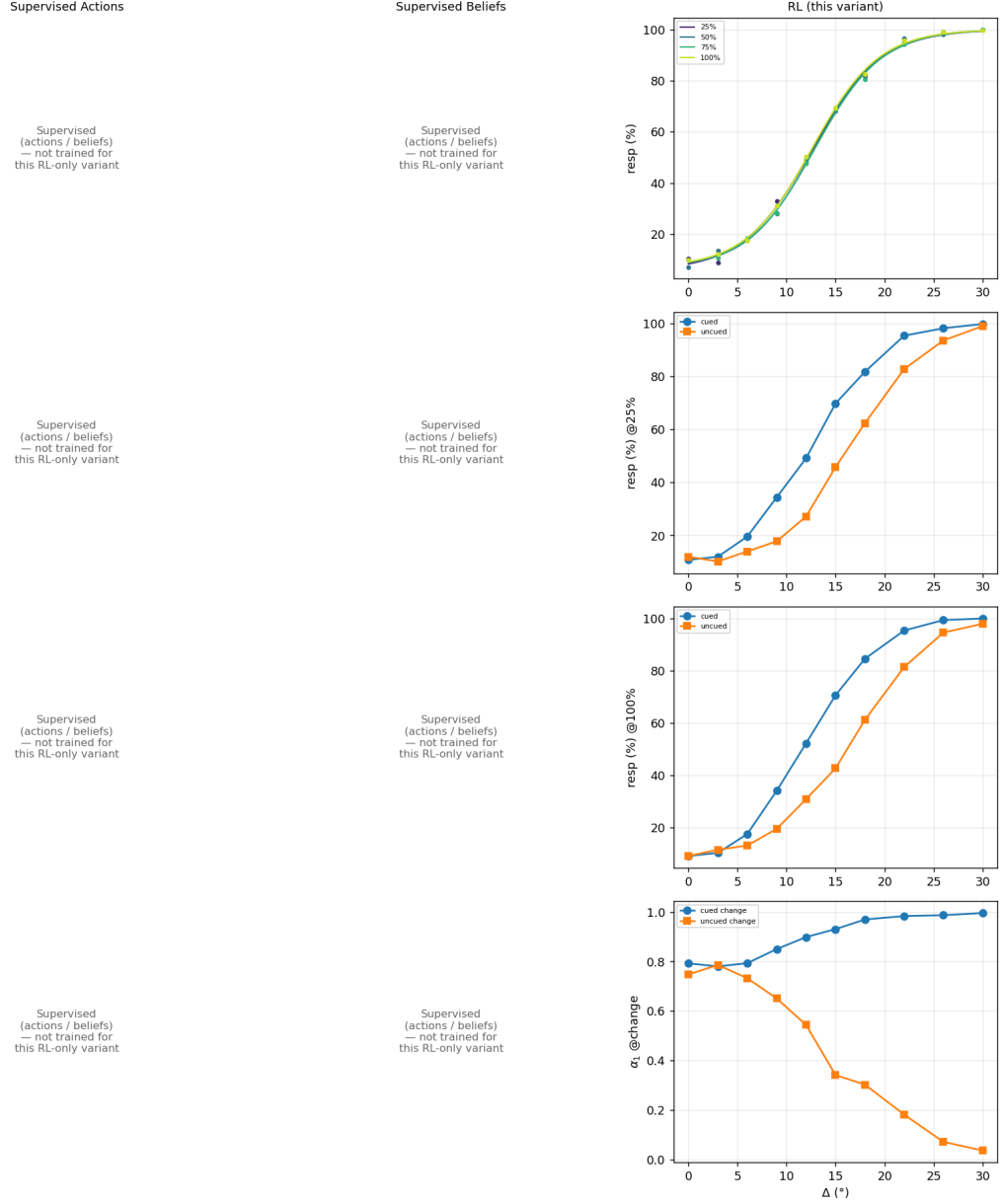


Figure 18: **Fig 17 analog — supervised vs. RL.** RL column reproduced (response by validity; cued vs. uncued at 25/100%; attention on S_1 at change vs. Δ); supervised columns are out-of-scope baselines for this RL-only variant.